GOLIAT: Configurable near- and far-field RF-EMF dosimetry studies in Sim4Life

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Summary

Computer models can estimate how Radiofrequency Electromagnetic Fields (RF-EMFs) deposit energy in the human body. A study may repeat the same model at many frequencies, device positions, incident directions, and polarizations. Each case needs a simulation scene, numerical grid, solver run, extracted result, and analysis record.

`GOLIAT` is an open-source Python workflow for these repeated studies in Sim4Life. Users describe a study in hierarchical JavaScript Object Notation (JSON) files. `GOLIAT` validates the merged settings and expands the study into individual jobs. It then sets up near-field device scenes or far-field incident-field scenes, runs the jobs locally or through oSPARC, extracts Specific Absorption Rate (SAR) and Absorbed Power Density (APD) results, and writes campaign tables and plots. Both exposure types have the same job identity and result layout.

Statement of need

Radiofrequency Electromagnetic Field (RF-EMF) dosimetry studies often require hundreds of Finite-Difference Time-Domain (FDTD) simulations (Hand, 2008; Hirata et al., 2021). Researchers must control the anatomical model, source, frequency, grid, normalization, and requested quantities in every case. A missed setting or changed file name makes direct comparison difficult. Large campaigns also need clear progress records and a way to rerun one failed phase without rebuilding the full study.

Sim4Life has anatomical models, electromagnetic solvers, dosimetry evaluators, and a Python Application Programming Interface (API) (Gosselin et al., 2014;

ZMT Zurich MedTech AG, 2026). Its interface has the operations for one simulation. However, a study still needs rules that list cases, set up exposure-specific scenes, name jobs, choose execution routes, check outputs, and combine results. A single study script can contain all these rules, but reviewers must then read program code to find the scientific choices.

`GOLIAT` stores the study choices in validated configuration files. The target users are dosimetry researchers, device developers, and test laboratories that run repeated Sim4Life studies. The configuration states the request. Each job directory records the run. Therefore, users can use one definition for setup, execution, extraction, and analysis.

Test laboratories can keep a configuration and its job records under change control for repeated pre-compliance studies. The laboratory still supplies qualified models, numerical validation, uncertainty analysis, release control, and its quality system.

State of the field

The closest alternative is a study-specific program built with the Sim4Life Python API. Through this interface, Sim4Life can create scenes, control the solver, sweep parameters, run batches, and extract results without the graphical interface (ZMT Zurich MedTech AG, 2026). A custom script can do all these tasks. However, the script must still define the study: case enumeration, device or incident-field scenes, normalization, job identity, result records, and campaign analysis. `GOLIAT` stores these recurring RF-EMF dosimetry choices in validated configurations and uses the same interfaces for near- and far-field studies.

Other public tools solve related problems. First, `openEMS` is a free equivalent-circuit FDTD solver with 1 g and 10 g Specific Absorption Rate (SAR) calculations (Liebig, 2026). It uses its own geometry, material, and result stack. Second, `PyAEDT` gives Python access to Ansys Electronics Desktop (Ansys, Inc., 2026). Users can automate its tasks through that interface. Third, `Snakemake` schedules file-based tasks across workstations and clusters (Köster & Rahmann, 2012). It can call a `GOLIAT` command, but it does not define a dosimetry case. These tools give users a solver, vendor API, or scheduler. They do not have GOLIAT's Sim4Life-specific study layer.

Therefore, `GOLIAT` is a configuration-to-results layer for Sim4Life. It uses local `iSolve` or oSPARC for the electromagnetic solve (IT'IS Foundation, 2026). It defines studies, placements, normalization, job records, result schemas, and analysis rules.

Software design

The design separates common study management from exposure-specific physics. The loader merges inherited files, checks the result, and creates the jobs. Near-field classes place device antenna models relative to anatomical landmarks and normalize results to input power. Far-field classes create incident plane waves or beamforming sources and normalize results to incident power density. Shared components run jobs, record status, track extraction, and analyze campaigns.

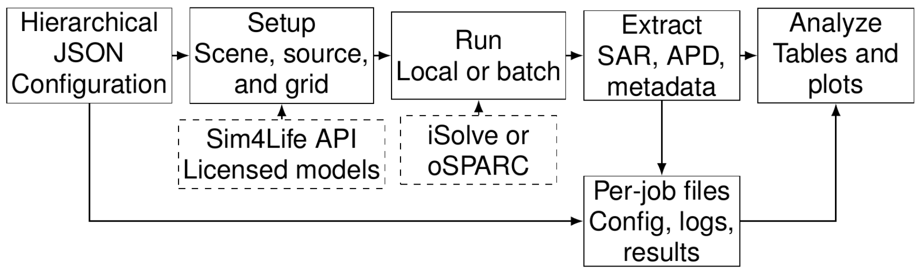

Figure 1: GOLIAT flowchart. The top row shows a study from hierarchical configuration to analysis. Dashed boxes mark licensed models and external execution services. Each simulation has a separate configuration, log, and result record.

Reviewers can read validated configuration files without tracing an unrestricted script. A base file has shared solver, material, and output settings. Smaller files set the phantoms, sources, frequencies, placements, directions, and polarizations for one study. `GOLIAT` writes the expanded configuration beside each result. Thus, users can inspect the exact input without reopening the base files.

Each job has a separate Sim4Life project and result directory. This gives more files than a single project. However, one solver failure affects one job, and operators can move jobs across workstations. Setup and extraction can use several Central Processing Unit (CPU) processes. Solver jobs can run through local `iSolve` calls, separate workers, or oSPARC batches. All execution routes use the same job state for progress logs and deliverable checks.

Extraction writes compact JavaScript Object Notation (JSON), serialized tissue tables, Hypertext Markup Language (HTML) reports, and simulation metadata from the Sim4Life results. Campaign analysis finds these records without reopening every project. It writes detailed and summary Comma-Separated Values (CSV) or Excel tables and groups plots by phantom, frequency, source, placement, and tissue. The documentation has installation guidance, five tutorials, configuration references, and an API reference (Wydaeghe, 2026a).

Research impact statement

`GOLIAT` was developed within the European GOLIAT project and has run two computational dosimetry campaigns. The near-field campaign has 352 jobs for two child anatomical models, eight frequencies, and 22 eye, cheek, and trunk placements. Every expected job directory has its expanded configuration, logs, and core SAR results.

The far-field campaign has 550 jobs for two adult and two child anatomical models at 15 frequencies between 450 MHz and 26 GHz. Its cases use environmental plane waves or auto-induced exposure. The *Physics in Medicine & Biology* article used the campaign's numerical data and tables (Wydaeghe et al., 2026a). Its postprocessed data and configuration family are deposited in Harvard Dataverse (Wydaeghe et al., 2026b). Together, the completed campaigns have 902 job records with the same configuration, execution, extraction, and analysis structure.

The software is distributed through the Python Package Index and has public documentation (Wydaeghe, 2026b, 2026a). Local scene construction and solver execution require a licensed Sim4Life installation and licensed anatomical models. The repository tests configuration handling, job management, and license-free analysis functions without those assets.

AI usage disclosure

During preparation of this work, the authors used Artificial Intelligence (AI) for language editing and preparation of the paper figure. The authors reviewed and edited the text and checked the software claims. The authors take full responsibility for the manuscript.

Conflict of interest

The authors declare no financial, personal, or professional conflict of interest related to this submission.

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